ECON 573 Problem Set 8 by Jimmy Zheng

Problem 1: ISL Chapter 6: Exercises 4

Exercise 4

a. (iii) the training RSS will increase steadily. As the value of λ increases, to minimize RSS + $\lambda \sum_{i=1}^{P} \beta_i^2$, the ridge regression shrinks all coefficients towards zero, which restricts the value that β_i can take, resulting in lower model flexibility. The decrease in model flexibility leads to an increase in training RSS.
b. (ii) is correct. When $\lambda = 0$, overly flexible OLS model has extremely high variance but low bias, resulting in overfitting and high test RSS. As the value of λ increases, more restrictions have been imposed on β_i's, resulting in substantial reduction in variance and mild increase in bias initially. However, when λ starts getting too large, the model become overly inflexible, the increase in bias outpaces the decrease in variance, and the test RSS starts increasing.
c. (iv) is correct. As we increase λ from 0, the constraints imposed on $\sum_{i=1}^{P} \beta_i^2$ become harsher, flexibility decrease, and the model contains smaller coefficients. Variance decreases because of lower model flexibility since the estimated parameters are now more robust against changes in the test datasets.
d. (iii) is correct. When $\lambda = 0$, no restrictions have been imposed on β_i's, resulting in highly flexible model with low bias. As we increase λ from 0, the constraints imposed on $\sum_{i=1}^{P} \beta_i^2$ become harsher, flexibility decrease, and bias increases.
e. (v) is correct. Irreducible errors are said to be irreducible because they cannot be predicted by those covariates X's and thus will NOT be affected the model flexibility (the value of λ). That is also the reason why the irreducible error will always provide an upper bound on the accuracy of our prediction for Y.

Problem 2: ISL Chapter 7. Exercise 3

```
X=seq(-2, 2, by=0.0001)
Y = 1 + X - 2*(X-1)^2*I(X>=1)

plot(X,Y, type = "l")
grid()
abline(h=0, lty = 2)
abline(v=0, lty = 2)
abline(v=1, lty = 2)
```

